

$$S_e \longrightarrow \boxed{e_s} \longrightarrow e$$

$$e = e_s$$

$$S_f \longrightarrow \boxed{f_s} \longrightarrow f$$

$$f = f_s$$

$$C \longleftarrow \begin{array}{c} \boxed{\frac{1}{C}} \longrightarrow e \\ \uparrow q \\ \boxed{\int} \longleftarrow f \end{array}$$

$$e = \frac{q}{C}$$

$$q = \int f dt + p(0)$$

$$C \longleftarrow \begin{array}{c} q = Ce \\ f = \frac{dq}{dt} \end{array}$$

$$\begin{array}{c} \boxed{C} \longleftarrow e \\ \uparrow q \\ \boxed{\frac{d}{dt}} \longrightarrow f \end{array}$$

$$I \longleftarrow \begin{array}{c} \boxed{\int} \longleftarrow e \\ \uparrow p \\ \boxed{\frac{1}{I}} \longrightarrow f \end{array}$$

$$f = \frac{p}{I}$$

$$p = \int e dt + p(0)$$

$$I \longleftarrow \begin{array}{c} p = If \\ e = \frac{dp}{dt} \end{array}$$

$$\begin{array}{c} \boxed{\frac{d}{dt}} \longrightarrow e \\ \uparrow f \\ \boxed{I} \longleftarrow f \end{array}$$

$$R \longleftarrow \begin{array}{c} \boxed{R} \longrightarrow e \\ \uparrow f \end{array}$$

$$e = Rf$$

$$\begin{array}{c} \boxed{R} \longrightarrow e \\ \uparrow f \end{array}$$

$$R \longleftarrow \begin{array}{c} \boxed{\frac{1}{R}} \longrightarrow f \\ \uparrow e \end{array}$$

$$e = \frac{f}{R}$$

$$\begin{array}{c} \boxed{\frac{1}{R}} \longrightarrow f \\ \uparrow e \end{array}$$

$$\begin{array}{c} e_1 \\ \hline f_1 \end{array} \xrightarrow{TF} \begin{array}{c} e_2 \\ \hline f_2 \end{array}$$

$$f_2 = n f_1$$

$$e_2 = n e_1$$

$$\begin{array}{c} \boxed{n} \\ \begin{array}{c} \longleftarrow e_1 \\ \longrightarrow e_2 \\ \longleftarrow f_1 \\ \longrightarrow f_2 \end{array} \end{array}$$

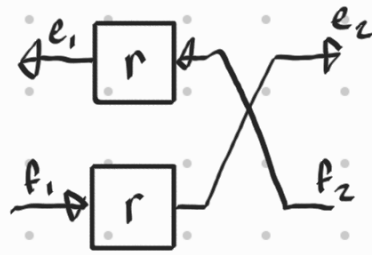
$$\begin{array}{c} \longrightarrow TF \longrightarrow \\ \ddot{n} \\ f_1 = \frac{f_2}{n} \\ e_1 = \frac{e_2}{n} \end{array}$$

$$\begin{array}{c} \boxed{\frac{1}{n}} \\ \begin{array}{c} \longrightarrow e_1 \\ \longrightarrow e_2 \\ \longleftarrow f_1 \\ \longleftarrow f_2 \end{array} \end{array}$$

$$\begin{matrix} e_1 \\ f_1 \end{matrix} \rightarrow \begin{matrix} GY \\ r \end{matrix} \begin{matrix} e_2 \\ f_2 \end{matrix}$$

$$e_2 = r f_1$$

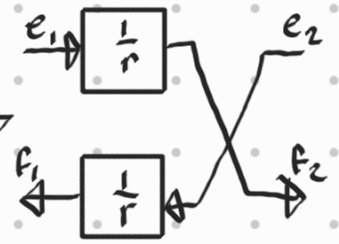
$$e_1 = r f_2$$



$$\begin{matrix} e_1 \\ f_1 \end{matrix} \rightarrow \begin{matrix} GY \\ r \end{matrix} \begin{matrix} e_2 \\ f_2 \end{matrix}$$

$$f_2 = \frac{e_1}{r}$$

$$f_1 = \frac{e_2}{r}$$



$$\frac{e_r}{f_e} \rightarrow RS \frac{T}{f_{s,irr}}$$

entropy

$$e_r = R \cdot f_r$$

$$f_{s,irr} = \frac{e_r f_r}{T} = \frac{R \cdot f_r^2}{T} \geq 0$$

$$T = \frac{e_r f_r}{f_{s,irr}}$$

$$\frac{T_{in}}{f_{sin}} \rightarrow RS \frac{T_{out}}{f_{sout}}$$

$$f_{sin} = \frac{dS_{in}}{dt} = \frac{1}{R} \frac{(T_{in} - T_{out})}{T_{in}}$$

$$f_{sout} = \frac{dS_{out}}{dt} = \frac{1}{R} \frac{(T_{in} - T_{out})}{T_{out}}$$

$$T_{so} = \frac{1}{R} \cdot \frac{\Delta T}{T_o}$$

$$\begin{pmatrix} e_1 \\ f_1 \end{pmatrix} = \begin{bmatrix} 0 & r \\ 1/r & 0 \end{bmatrix} \begin{pmatrix} e_2 \\ f_2 \end{pmatrix}$$